

On the Brown–Proschan model when repair effects are unknown

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A device is repaired after failure. The Brown–Proschan (BP) model assumes that the repair is perfect with probability p and minimal with probability $(1-p)$. Theoretical results usually suppose that each repair effect (perfect or minimal repair) is known. However, this is not generally the case in practice. In this paper, we study the behavior of the BP model when repair effects are unknown. In this context, the main features of the failure process are derived: distribution functions of times between failures, failure intensity, likelihood function, etc. We propose to estimate the repair efficiency parameter p and the parameters of the first time to failure distribution with the likelihood function or equivalently the EM algorithm. We also propose to combine a moment estimation of the scale parameter and a maximum likelihood estimation of other parameters. Copyright © 2010 John Wiley & Sons, Ltd.

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1. Introduction

All major industrial systems are subjected to corrective maintenance actions or repairs. The basic assumptions concerning repair efficiency are known as minimal repair or as bad as old (ABAO) and perfect repair or as good as new (AGAN). The corresponding random processes are non-homogeneous Poisson processes (NHPP) and renewal processes (RP), respectively. More realistic notions of repair fall between these two extremes. Many models have already been proposed for imperfect repair effects (see for example a review in [1]). One of the most widely used is the Brown–Proschan (BP) model [2], in which system state after repair is assumed to be AGAN with probability p and ABAO with probability $1-p$.

Whitaker and Samaniego [3] have highlighted an identifiability problem for p . When the first time to failure is exponentially distributed, distributions of times between failures do not depend on the value of p . In this case, repairs are unnecessary since the system does not wear-out. The failure process behavior does not depend on repair efficiency, i.e. p . If the first time to failure is assumed to be non-exponentially distributed, p should be identifiable.

Authors have usually assumed that the effect of each repair (ABAO or AGAN) is known, perhaps because of this identifiability problem, and have developed statistical methods for first time to failure distribution (representing wear-out of the new system in the absence of repair). Whitaker and Samaniego [3] thus proposed a kind of non-parametric Kaplan–Meier estimator. Hollander *et al.* [4] and later Kvam *et al.* [5] also developed non-parametric estimators but for a generalization of the BP model proposed by Block, Borges and Savits (BBS) [6] in which the probability of perfect repair depends on repair time. Bathe and Franz [7] studied maximum likelihood (ML) parametric estimators for another generalization of the BP model including possible partial repair effects and random repair action duration. For the BBS model, Agustin and Peña [8] proposed a goodness-of-fit test for first time to failure distribution when repair effects are known.

However, in practice, repair effects are generally unknown, especially since maintenance is not AGAN nor ABAO: parameter p only represents the degree of efficiency of repair actions. To our knowledge, only four papers deal with the BP model with unknown repair effects. Lim [9] estimated the parameters for the first time to failure distribution and repair efficiency with the Expectation-Maximization (EM) algorithm. Lim and Lie [10] used an SEM algorithm to estimate the parameters of a generalization of the BP model that allows first-order dependency between two consecutive repair effects, but they assumed that only some repair effects were unknown. Lim, Lu and Park [11] proposed another method based on Bayesian analysis: they assumed a prior beta distribution for parameter p . Langseth and Lindqvist [12] generalized the BP

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model for imperfect preventive maintenances. They proposed to estimate the parameters of the model (including the repair efficiency parameter) with the likelihood function.

The aim of this paper is to study the BP model when repair effects are unknown. First, the notations are defined and identifiability of the model is proved. In Section 3, we highlight the specific behavior of the failure process when repair effects are unknown. In Section 4, failure process behavior is characterized using its likelihood function or failure intensity. In Section 5, we develop two methods to compute ML estimators (MLEs). The first consists a direct likelihood maximization; the second is based on the EM algorithm. We also propose to combine moment estimation of the scale parameter with ML estimation of other parameters. Finally, in Section 6, these methods are applied to Kumar and Klefsjö [13]'s load-haul-dump (LHD) machines data set, and the empirical bias and standard deviation of the BP parameters estimators are computed numerically.

2. BP model with unknown repair effects

2.1. Notations

Let $0 = T_0 < T_1 < \dots$ be the successive failure times of a repairable system. The cumulative number of failures observed up to t is denoted by $N_t = \sum_{i=1}^{+\infty} \mathbb{1}_{\{T_i \leq t\}}$ and the time between failures is $X_i = T_{i+1} - T_i$ for $i \geq 1$. A repair is assumed to be performed after each failure and the duration of repair action is not taken into account. The i th repair effect is assumed to depend on external factors. These external factors are represented by random variables $B = \{B_i\}_{i \geq 1}$, called external variables. The B variables are assumed to be mutually independent and independent of previously observed repair times ($B_i \perp T_j$ for $i \geq j$).

The failure process can be defined by its failure intensity,

$$\forall t \geq 0, \quad \lambda_t(N, B) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P(N_{t+\Delta t} - N_t = 1 | \mathcal{H}_{t-})$$

where N_{t-} is the left-hand limit of N_t and $\mathcal{H}_{t-} = \sigma(\{N_s, B_{N_s}\}_{0 \leq s < t})$ is the history of the failure process. When no external variables are considered, the failure process is called a self-excited point process. If, in addition, $\lambda_t = \lim_{\Delta t \rightarrow 0} (1/\Delta t) P(N_{t+\Delta t} - N_t = 1 | N_{t-}, \underline{T}_{N_{t-}})$ (where $\underline{T}_n = (T_1, \dots, T_n)$) is assumed to be left-continuous with right-hand limits, then it can be shown [14] that λ_t is unique (up to indistinguishability) and characterizes the failure process completely. Λ_t denotes the cumulative failure intensity, i.e. the integral of the failure intensity over $[0, t]$.

In a parametric approach, failure intensity is assumed to depend on certain parameters denoted by θ . These parameters can be estimated using the ML method. For a single observation $\underline{t}_{n_t}, n_t$ of the failure process over a time interval $[0, t]$ (where t may or may not be a failure time), the likelihood function verifies [14]:

$$\begin{aligned} L_t(\theta) &= f_{\underline{T}_{n_t} | N_t = n_t}(\underline{t}_{n_t}) P(N_t = n_t) \\ L_t(\theta) &= \left[\prod_{i=1}^{n_t} \lambda_{t_i} \right] \exp(-\Lambda_t) \end{aligned} \quad (1)$$

where $f_{X|A}(x)$ is the conditional density of X given A . It is equal to $\lim_{\Delta x \rightarrow 0} 1/\Delta x P(x < X \leq x + \Delta x | A)$. The MLE is the value of the parameters that maximizes the likelihood or, alternatively, the logarithm of the likelihood, called log-likelihood.

$$\hat{\theta} = \arg \max_{\theta} L_t(\theta) = \arg \max_{\theta} \log(L_t(\theta))$$

2.2. BP model definition

Before the first failure, failure intensity is assumed to be a deterministic and continuous function of time, $\lambda(t)$, called initial intensity. The initial intensity is the hazard rate of the first time to failure. It represents the system's intrinsic wear-out. It is assumed to be an increasing not-always-null function of time, since the system is assumed to wear-out in the absence of repair. Its integral over $[0, t]$ is denoted by $\Lambda(t)$. For example, we can assume that the first time to failure is Weibull-distributed:

$$\lambda(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1} \quad \text{and} \quad \Lambda(t) = \left(\frac{t}{\eta} \right)^{\beta} \quad \text{with } \eta \in]0, +\infty[\text{ and } \beta \in [1, +\infty[\quad (2)$$

η is the scale parameter and β the shape parameter.

The BP model assumptions are:

- With probability p , system state after a repair is AGAN, i.e. failure intensity is the same as for a new system.
- With probability $1 - p$, system state after a repair is ABAO, i.e. failure intensity is not affected by the repair action.
- The repair effects (AGAN or ABAO) are mutually independent, and independent of already observed failure times.

The random effect of repair actions can be represented by the variables B :

$$B_n = \begin{cases} 1 & \text{if the repair is AGAN} \\ 0 & \text{if the repair is ABAO} \end{cases}$$

Previous assumptions imply that external variables verify:

$$\forall n \geq 1, \quad P(B_n = 1 | \underline{T}_n, \underline{B}_{n-1}) = P(B_n = 1) = p$$

Let $B_0 = 1$. For $n \geq 0$, let A_n be the time elapsed between T_n and the last previous perfect repair T_k , with $k = \max\{i < n | B_i = 1\}$. A_n is called effective age and verifies:

$$\begin{aligned} A_0 &= 0 \\ A_n &= (1 - B_n)[X_n + A_{n-1}] \quad \text{for } n \geq 1 \end{aligned}$$

Let $\sum_{j=k}^l \cdot = 0$ and $\prod_{j=k}^l \cdot = 1$ for $k > l$. Then

$$\forall n \geq 0, \quad A_n = \sum_{j=1}^n \left[\prod_{k=j}^n (1 - B_k) \right] X_j$$

At any time t , the system is equivalent to new system that has been in operation for the time elapsed since the last perfect repair: $t - T_{N_{t-}} + A_{N_{t-}}$. Therefore, its failure intensity verifies:

$$\lambda_t(N, B) = \lambda \left(t - T_{N_{t-}} + \sum_{j=1}^{N_{t-}} \left[\prod_{k=j}^{N_{t-}} (1 - B_k) \right] X_j \right)$$

This expression remains true whatever be the assumptions concerning repair effects B , and is also the failure intensity of the more general Kijima type II model [15].

Let B_i^j denotes the event: at time T_i , the last perfect repair is the j th previous one $B_i^j = \{B_{i-j} = 1, \{B_{i-k} = 0\}_{0 \leq k < j}\}$. $\{B_{N_{t-}}^j\}_{0 \leq j \leq N_{t-}}$ is a partition of the probability space. In addition, at time t , if $\mathbb{1}_{\{B_{N_{t-}}^j\}} = 1$ then the last perfect repair is $T_{N_{t-}-j}$; otherwise $\mathbb{1}_{\{B_{N_{t-}}^j\}} = 0$. Hence, the failure intensity also verifies:

$$\lambda_t(N, B) = \sum_{j=0}^{N_{t-}} \mathbb{1}_{\{B_{N_{t-}}^j\}} \lambda(t - T_{N_{t-}-j}) = \prod_{j=0}^{N_{t-}} [\lambda(t - T_{N_{t-}-j})]^{\mathbb{1}_{\{B_{N_{t-}}^j\}}}$$

2.3. Identifiability

In statistics, identifiability is a property which a model must satisfy in order for inference to be possible. A model is said to be identifiable if distinct values of the parameters correspond to distinct probability distributions of the observations. In this section, we will prove that the BP model remains identifiable even if the repair effects are not observed.

The initial intensity $\lambda(t)$ is identifiable because it is the hazard rate of T_1 . To prove the identifiability of p , the joint density of the two first times between failures is calculated.

$$\begin{aligned} f_{X_1, X_2}(x_1, x_2) &= p f_{X_1, X_2 | B_1=1}(x_1, x_2) + (1 - p) f_{X_1, X_2 | B_1=0}(x_1, x_2) \\ &= p f_{X_2 | B_1=1, X_1=x_1}(x_2) f_{X_1}(x_1) + (1 - p) f_{X_2 | B_1=0, X_1=x_1}(x_2) f_{X_1}(x_1) \end{aligned}$$

However,

$$\begin{aligned} f_{X_2 | B_1=1, X_1=x_1}(x_2) &= f_{X_1}(x_2) = \lambda(x_2) \exp(-\Lambda(x_2)) \\ f_{X_2 | B_1=0, X_1=x_1}(x_2) &= \lambda(x_2 + x_1) \exp(-(\Lambda(x_2 + x_1) - \Lambda(x_1))) \end{aligned}$$

Therefore,

$$f_{X_1, X_2}(x_1, x_2) = \lambda(x_1)\lambda(x_1 + x_2)\exp(-\Lambda(x_1 + x_2)) + p\lambda(x_1)[\lambda(x_2)\exp(-\Lambda(x_1) - \Lambda(x_2)) - \lambda(x_1 + x_2)\exp(-\Lambda(x_1 + x_2))]$$

Hence, p is not identifiable if and only if, for all $x_1 \geq 0$ and $x_2 \geq 0$,

$$\lambda(x_2)\exp(-\Lambda(x_1) - \Lambda(x_2)) = \lambda(x_1 + x_2)\exp(-\Lambda(x_1 + x_2))$$

We take the anti-derivatives with respect to x_2 on both sides of this equation, and prove that it is equivalent to:

$$\Lambda(x_1) + \Lambda(x_2) = \Lambda(x_1 + x_2)$$

Therefore, if the first time to failure is assumed to be non-exponentially distributed, the parameter p of the model is identifiable.

3. Failure process behavior

In order to analyze the behavior of the failure process when repair effects are unknown, we study the effective ages and times between failures distributions. These distributions do not characterize the failure process completely, because they are not conditional to the failure process history. However, they do highlight changes induced by the fact that the external process B is hidden.

3.1. Effective age

All the results of this section and the next are based on the theorem below which gives the successive distributions of the effective ages. For any random variable Z , let $F_Z(z) = P(Z \leq z)$ and $S_Z(z) = P(Z > z)$ denote its distribution function and survival function, respectively.

Theorem 1

$$F_{A_n}(x) = 1 - (1 - p) \exp(-\Lambda(x)) \left[\sum_{k=0}^{n-1} (1 - p)^k \frac{\Lambda^k(x)}{k!} \right] \quad (3)$$

Proof

The distribution functions of the effective ages verify: $F_{A_n}(x) = \sum_{i=0}^n P(A_n \leq x | B_n^i) P(B_n^i)$. First, for $i < n$, $P(B_n^i) = p(1 - p)^i$, and for $i = n$, $P(B_n^n) = (1 - p)^n$, since B_0 is assumed to be equal to one. Second, for $i = 0$,

$$P(A_n \leq x | B_n^0) = P(A_n \leq x | B_n = 1) = P(0 \leq x) = 1$$

Let $\{\tilde{N}_t\}_{t \geq 0}$, $\{\tilde{T}_i\}_{i \geq 1}$, $\{\tilde{X}_i\}_{i \geq 1}$ denote (respectively) counting process, failure times and times between failures of an NHPP with a failure intensity $\lambda(t)$. Using the traditional results of this kind of process, it can be proved, for $i > 0$:

$$\begin{aligned} P(A_n \leq x | B_n^i) &= P(\tilde{X}_1 + \dots + \tilde{X}_i \leq x) = P(\tilde{T}_i \leq x) = 1 - P(\tilde{N}_x < i) \\ &= 1 - \sum_{k=0}^{i-1} \frac{\Lambda^k(x)}{k!} e^{-\Lambda(x)} \end{aligned}$$

Finally, the distribution functions of the effective ages verify:

$$\begin{aligned} F_{A_n}(x) &= p + p \left[\sum_{i=1}^{n-1} (1 - p)^i \right] + (1 - p)^n - p e^{-\Lambda(x)} \left[\sum_{i=1}^{n-1} \sum_{k=0}^{i-1} (1 - p)^i \frac{\Lambda^k(x)}{k!} \right] - (1 - p)^n e^{-\Lambda(x)} \left[\sum_{k=0}^{n-1} \frac{\Lambda^k(x)}{k!} \right] \\ F_{A_n}(x) &= p + p(1 - p) \frac{1 - (1 - p)^{n-1}}{1 - (1 - p)} + (1 - p)^n - p e^{-\Lambda(x)} \left[\sum_{k=0}^{n-2} \sum_{i=k+1}^{n-1} (1 - p)^i \frac{\Lambda^k(x)}{k!} \right] - (1 - p)^n e^{-\Lambda(x)} \left[\sum_{k=0}^{n-1} \frac{\Lambda^k(x)}{k!} \right] \\ &= 1 - p e^{-\Lambda(x)} \left[\sum_{k=0}^{n-2} \frac{\Lambda^k(x)}{k!} (1 - p)^{k+1} \frac{1 - (1 - p)^{n-k-1}}{1 - (1 - p)} \right] - (1 - p)^n e^{-\Lambda(x)} \left[\sum_{k=0}^{n-1} \frac{\Lambda^k(x)}{k!} \right] \\ &= 1 - (1 - p) e^{-\Lambda(x)} \left[\sum_{k=0}^{n-1} (1 - p)^k \frac{\Lambda^k(x)}{k!} \right] \end{aligned}$$

□

The first obvious consequence of this theorem is that the effective ages converge in law. In order to have a bounded limit, p is assumed to be non-null. A similar convergence result has already been proved by Last and Szekli [16] for a larger class of imperfect repair model (a generalization of the Kijima type II model) including the BP model.

Corollary 1

If $p > 0$, $A_{n+1} \xrightarrow{\mathcal{L}} A_\infty$, where $F_{A_\infty}(x) = 1 - (1 - p) \exp(-p\Lambda(x))$.

Last and Szekli also proved other asymptotic criteria, including the convergence of the expectation of A_{n+1} . This result can also be deduced from Theorem 1. Here, however, we are able to compute not only the limit, but all the values for $n \geq 1$ of the expectation of A_{n+1} .

Corollary 2

If $\varepsilon > 0$ exists such that $x^{1+\varepsilon} \exp(-\Lambda(x)) = o(1)$, then

$$E[A_n] = (1 - p) \int_0^{+\infty} \exp(-\Lambda(x)) \left[\sum_{k=0}^{n-1} (1 - p)^k \frac{\Lambda^k(x)}{k!} \right] dx \quad (4)$$

And if in addition $p > 0$, then

$$E[A_\infty] = \lim_{n \rightarrow \infty} E[A_n] = (1 - p) \int_0^{+\infty} \exp(-p\Lambda(x)) dx \quad (5)$$

Proof

The corollary is directly proved by applying the well-known formula $E[Z] = \int_0^\infty S_Z(z) dz$; however, this formula only remains true if $z S_Z(z)$ vanishes as $z \rightarrow \infty$. The assumption before Equation (4) ensures that this condition is verified. $\square$

3.2. Time between failures

The distribution of the time between failures can be deduced from the effective age distributions.

Corollary 3

For $n \geq 1$,

$$S_{X_{n+1}}(x) = (1 - p)^n \int_0^{+\infty} \lambda(t) \frac{\Lambda^{n-1}(t)}{(n-1)!} e^{-\Lambda(x+t)} dt + p \left[\sum_{i=0}^{n-1} (1 - p)^i \int_0^{+\infty} \lambda(t) \frac{\Lambda^{i-1}(t)}{(i-1)!} e^{-\Lambda(x+t)} dt \right] + p e^{-\Lambda(x)}$$

And, if $p > 0$, then $X_n \xrightarrow{\mathcal{L}} X_\infty$, where

$$S_{X_\infty}(x) = p \int_0^{+\infty} \lambda(x+v) \exp(-\Lambda(x+v) + (1-p)\Lambda(v)) dv$$

Proof

As for the proof of Theorem 1, the time between failures distribution function verifies $S_{X_{n+1}}(x) = \sum_{i=0}^n P(X_{n+1} \geq x | B_n^i) P(B_n^i)$ and

$$P(X_{n+1} \geq x | B_n^i) = P(\tilde{X}_{i+1} \geq x) = \begin{cases} \exp(-\Lambda(x)) & \text{if } i = 0 \\ \int_0^{+\infty} e^{-(\Lambda(x+t) - \Lambda(t))} f_{\tilde{T}_i}(t) dt & \text{otherwise} \end{cases}$$

Finally, the failure time density of an NHPP verifies $f_{\tilde{T}_i}(t) = \lambda(t) (\Lambda^{i-1}(t) / (i-1)!) e^{-\Lambda(t)}$.

To prove the convergence result, $\exp((1-p)\Lambda(v))$ is replaced by its Taylor series. Then, times between failures converge in law to the random variable X_∞ with the survival function:

$$\begin{aligned} S_{X_\infty}(x) &= p e^{-\Lambda(x)} - p(1-p) \int_0^{+\infty} \lambda(v) e^{-\Lambda(x+v)+(1-p)\Lambda(v)} dv \\ &= p e^{-\Lambda(x)} + p \left([e^{-\Lambda(x+v)+(1-p)\Lambda(v)}]_0^\infty + \int_0^{+\infty} \lambda(x+v) e^{-\Lambda(x+v)+(1-p)\Lambda(v)} dv \right) \\ &= p \int_0^{+\infty} \lambda(x+v) e^{-\Lambda(x+v)+(1-p)\Lambda(v)} dv \end{aligned}$$

□

Finally, the following result concerning the time between failures expectation can be deduced from Corollary 2. In fact,

$$E[A_n] = E[(1 - B_n)(X_n + A_{n-1})]$$

And since B_n is independent of X_n and A_{n-1} , the previous equation implies that:

$$E[X_n] = \frac{E[A_n]}{1-p} - E[A_{n-1}]$$

Corollary 4

If $\varepsilon > 0$ exists such that $x^{1+\varepsilon} \exp(-\Lambda(x)) = o(1)$, then

$$E[X_n] = \int_0^{+\infty} \exp(-\Lambda(x)) \left[(1-p)^{n-1} \frac{\Lambda^{n-1}(x)}{(n-1)!} + p \sum_{k=0}^{n-2} (1-p)^k \frac{\Lambda^k(x)}{(k)!} \right] dx$$

And if in addition $p > 0$, then

$$E[X_\infty] = \lim_{n \rightarrow \infty} E[X_n] = p \int_0^{+\infty} \exp(-p\Lambda(x)) dx$$

If the first time to failure is Weibull-distributed, these expressions can be simplified using the Gamma function: $\Gamma(x) = \int_0^{+\infty} u^{x-1} e^{-u} du$.

Corollary 5

If the first time to failure is Weibull-distributed (2), then

$$E[X_n] = \frac{\eta}{\beta} \left[(1-p)^{n-1} \frac{\Gamma(n-1+1/\beta)}{\Gamma(n)} + p \left[\sum_{k=0}^{n-2} (1-p)^k \frac{\Gamma(k+1/\beta)}{\Gamma(k+1)} \right] \right]$$

And if in addition $p > 0$, then

$$E[X_\infty] = \frac{p^{(\beta-1)/\beta} \eta}{\beta} \Gamma\left(\frac{1}{\beta}\right) = p^{1-1/\beta} E[X_1]$$

The more important result of this section is Corollary 3: it explains the distributions of the successive times between failures. Figure 1 shows the hazard rate of these times between failures when the first time to failure is Weibull-distributed. This hazard rate verifies $\lambda_{X_n}(x) = S'_{X_n}(x)/S_{X_n}(x)$ or alternatively,

$$\begin{aligned} \lambda_{X_n}(x) &= \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P(x < X_n \leq x + \Delta x | X_n > x) \\ &= \int_0^{+\infty} \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P(x < X_n \leq x + \Delta x | A_n = v, X_n > x) f_{A_n | X_n > x}(v) dv \\ &= \int_0^{+\infty} \lambda(x+v) f_{A_n | X_n > x}(v) dv \end{aligned}$$

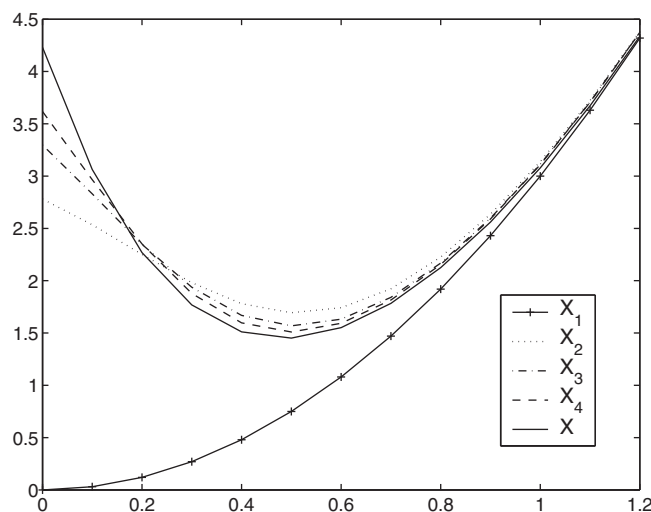


Figure 1. Hazard rate of the successive times between failures for $p=0.3$.

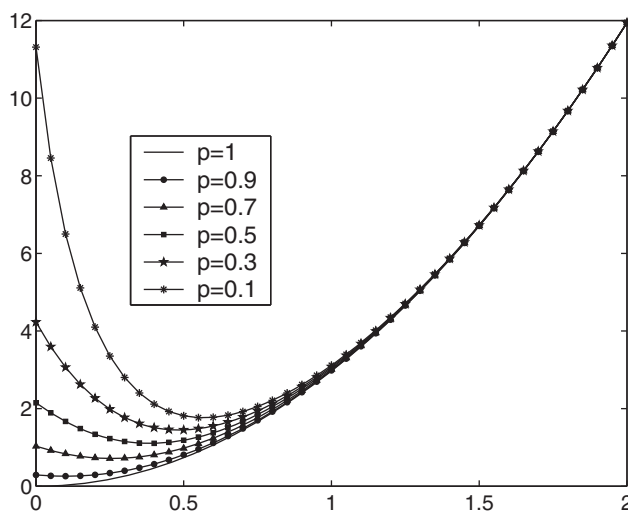


Figure 2. Hazard rate of X_∞ function of p .

It can be observed that after a repair, the system has a hazard rate bathtub curve. This shape is due to unknown repair effects or equivalently due to the dependency between A_n and $X_n > x$:

- First, the initial value of the hazard rate grows as n increases. In fact, $f_{A_n|X_n>0}(v) = f_{A_n}(v)$ and then $\lambda_{X_n}(0) = E[\lambda(A_n)]$. Theorem 1 shows that for all $x \geq 0$, $F_{A_{n+1}}(x) \leq F_{A_n}(x)$, i.e. effective ages grow stochastically. Then, if initial intensity is assumed to be an increasing function, $E[\lambda(A_n)] \leq E[\lambda(A_{n+1})]$.
- Second, if the system survives for a sufficiently long time after a repair, the last repair must have been a perfect repair. Then, for a large enough x , $f_{A_n|X_n>x}$ remains equivalent to a Dirac function in 0, and $\lambda_{X_n}(x) \sim \lambda(x)$.

In practice, for large systems, the same phenomenon is observed: they are more inclined to break down just after repair because of mishandling.

Figure 2 shows, for the same initial intensity, the influence of the repair efficiency parameter p on the shape of the hazard rate of the asymptotic time between failures X_∞ . As p decreases, the initial value of the hazard rate increases. In fact, the probability of poor repair becomes higher.

3.3. A kind of asymptotic renewal property

In the previous section, we noticed that if the time elapsed between two repairs is sufficiently long, the failure intensity becomes equal to the initial intensity. In this section, a similar property is proved but for conditional survival functions.

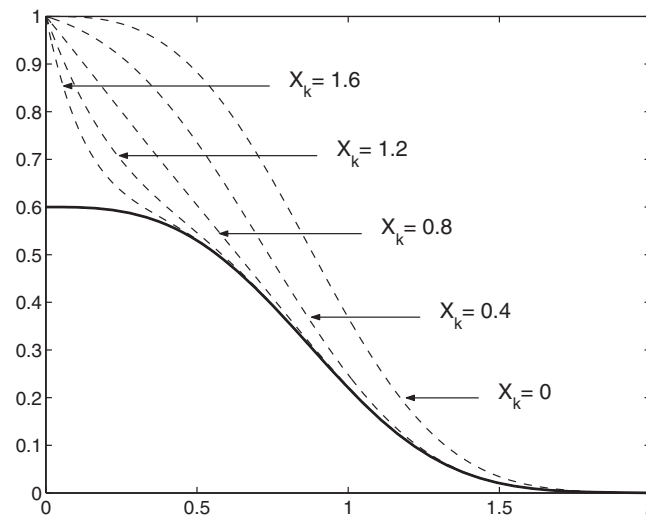


Figure 3. Bounds for the conditional survival function of X_{k+1} .

Theorem 2

If the initial intensity is a strictly increasing function, then for $k \geq 1$:

$$0 \leq P(X_{k+1} > x | X_k > y, \underline{X}_{k-1}) - p e^{-\Lambda(x)} \leq (1-p) \exp(-\Lambda(x+y) + \Lambda(y))$$

Proof

$$P(X_{k+1} > x | X_k > y, \underline{X}_{k-1}) = P(B_k = 1) P(X_{k+1} > x | B_k = 1) + P(B_k = 0) P(X_{k+1} > x | B_k = 0, X_k > y, \underline{X}_{k-1})$$

However, $P(X_{k+1} > x | B_k = 1) = \exp(-\Lambda(x))$ and

$$\begin{aligned} P(X_{k+1} > x | B_k = 0, X_k > y, \underline{X}_{k-1}) &= E[X_{k+1} > x | B_k = 0, X_k > y, \underline{X}_{k-1}] \\ &= E[E[X_{k+1} > x | A_k] | B_k = 0, X_k > y, \underline{X}_{k-1}] \\ P(X_{k+1} > x | A_k) &= \exp(-\Lambda(x + A_k) + \Lambda(A_k)) \end{aligned}$$

Since λ is an increasing function, $g_x(y) = \exp(-\Lambda(x+y) + \Lambda(y))$ is a decreasing function and then:

$$P(X_{k+1} > x | B_k = 0, X_k > y, \underline{X}_{k-1}) \leq e^{-\Lambda(x+y) + \Lambda(y)}$$

□

For $p=0.6$ and $\lambda(t)=3t^2$, Figure 3 shows the function $p \exp(-\Lambda(x))$ as a solid line and $p \exp(-\Lambda(x)) + (1-p) \exp(-\Lambda(x+y) + \Lambda(y))$ as a dashed line for different values of y . Theorem 2 means that the conditional survival functions of the times between failures take their values between these two bounds, i.e.

$$P(X_{k+1} > x | \underline{X}_k) = p e^{-\Lambda(x)} + g_{\underline{X}_k}(x)$$

with $0 \leq g_{\underline{X}_k}(x) \leq (1-p) \exp(-\Lambda(x + X_k) + \Lambda(X_k))$. However, when $x \rightarrow \infty$, $(1-p) \exp(-\Lambda(x + X_k) + \Lambda(X_k))$ vanishes and

$$P(X_{k+1} > x | \underline{X}_k) \sim p e^{-\Lambda(x)}$$

If repair actions are not always minimal ($p > 0$), the tails of all the conditional distributions of times between failures are the same and do not depend on previous failure times. They are the tails of a RP with time between failures having survival function $p \exp(-\Lambda(x))$. Furthermore, if the previous time between failures is large enough, $p \exp(-\Lambda(x))$ is also a good approximation of the conditional distribution of the times between failures for all values of x except the very small ones.

4. Failure process characterization

In this section, the failure process of the BP model with unknown repair effects is characterized by its failure intensity and two computable expressions of the likelihood function are provided in order to be able to estimate the parameters of the model.

4.1. Innovation theorem

Even if a failure process is a function of an external process, it can be viewed as a self-excited point process. The connection between the failure intensities $\lambda_t(N, B)$ and λ_t is made using the innovation theorem [14].

Theorem 3

Let $\mathcal{H}_t^1$ and $\mathcal{H}_t^2$ be two filtrations with, for all $t \geq 0$, $\mathcal{H}_t^2 \subseteq \mathcal{H}_t^1$. Consider a counting process N adapted to both filtrations. Suppose N has a failure intensity $\lambda_t(\mathcal{H}^1)$ with respect to the larger filtration. Then, with respect to the smaller filtration, N has a failure intensity $\lambda_t(\mathcal{H}^2)$ that verifies: $\lambda_t(\mathcal{H}^2) = E[\lambda_t(\mathcal{H}^1) | \mathcal{H}_{t-}^2]$, where $\mathcal{H}_{t-}^2 = \bigcup_{s < t} \mathcal{H}_s^2$.

In the case of the BP model, this theorem implies: $\lambda_t = E[\lambda_t(N, B) | N_{t-}, \underline{T}_{N_{t-}}]$. λ_t is called the unconditional failure intensity of the system, in opposition to $\lambda_t(N, B)$, which is called the conditional (to the external process) failure intensity. We refer to the unconditional failure intensity simply as the failure intensity. The advantage of unconditional failure intensity is that it characterizes the failure process as a self-excited point process. Thus, all the results developed for this kind of process can be applied. In particular, λ_t characterizes the failure process completely.

Failure intensity can be directly deduced from the innovation theorem, but proving it directly with the innovation theorem is rather technical. Hereafter, we propose to deduce the failure intensity from the likelihood function, which is much easier.

4.2. Likelihood function

Let us denote:

$$f_{\underline{T}_{n_t}, N_t = n_t}(\underline{t}_{n_t}) = f_{\underline{T}_{n_t} | N_t = n_t}(\underline{t}_{n_t}) P(N_t = n_t)$$

The parameters of the BP model are:

- p , which characterizes the repair efficiency and
- the parameters of the initial intensity, denoted by λ , which characterize the behavior of a new system in the absence of repair. If the first time to failure is Weibull-distributed (2), then $\lambda = (\eta, \beta)$.

These parameters, $\theta = (p, \lambda)$, can be estimated using the ML method. For the BP model, we propose two computable expressions of the likelihood function with recursive formulas.

Theorem 4

For a single observation of the failure process over $[0, t]$ (t may or may not be a failure time), the likelihood function verifies

$$L_t(\theta) = \sum_{j=0}^{n_t-} p^{\mathbb{1}_{\{j \neq 0\}}} (1-p)^{n_t-j} \left[\prod_{i=j+1}^{n_t} \lambda(t_i - t_j) \right] e^{-\Lambda(t-t_j)} L_{t_j}(\theta)$$

with $L_0(\theta) = 1$.

Proof

The likelihood function verifies:

$$L_t(\theta) = f_{\underline{T}_{n_t}, N_t = n_t}(\underline{t}_{n_t}) = \sum_{j=0}^{+\infty} f_{\underline{T}_{n_t}, N_t = n_t | B_{n_t-}^j}(\underline{t}_{n_t}) P(B_{n_t-}^j) \quad (6)$$

First, for $j > n_t$, $P(B_{n_t-}^j) = 0$, since B_0 is assumed to be equal to 1. Then in (6), the sum has no infinite term and stops at $j = n_t$.

Second, for $j = n_t$, $P(B_{n_t-}^j) = (1-p)^{n_t}$ and given $B_{n_t-}^j$, the failure process is equivalent to an NHPP. It is well known that the joint density of the observations verifies:

$$f_{\underline{T}_{n_t}, N_t = n_t | B_{n_t-}^j}(\underline{t}_{n_t}) = \left[\prod_{i=1}^{n_t} \lambda(t_i) \right] e^{-\Lambda(t)}$$

Finally, for $j < n_t$, $P(B_{n_t-j}^j) = p(1-p)^j$ and given $B_{n_t-j}^j$, the failure process has been renewed in t_{n_t-j} . It can be viewed as two independent processes. Before t_{n_t-j} , the failure process is that of the BP model, with censorship in t_{n_t-j} . The joint density of the observation is then equal to the likelihood function $L_{t_{n_t-j}}(\theta)$. After t_{n_t-j} , the failure process is equivalent to an NHPP initialized at t_{n_t-j} . The joint density of the observations verifies:

$$f_{\underline{T}_{n_t}, N_t=n_t | B_{n_t-j}^j}(\underline{t}_{n_t}) = \left[\prod_{i=1}^{n_t} \lambda(t_i - t_{n_t-j}) \right] e^{-\Lambda(t-t_{n_t-j})} L_{t_{n_t-j}}(\theta)$$

The theorem is proved by using these results in (6) and by changing the index of the sum from j to $k = n_t - j$. $\square$

For $i \leq n_t$, let us define the BP likelihood function, over a time interval $[0, t - t_i]$, associated with $(t_{i+1} - t_i, \dots, t_{n_t} - t_i, n_t - i)$ the positive part of the translation of the observations of $-t_i$:

$$L_{i,t}(\theta) = f_{\underline{T}_{n_t-i}, N_t-i=n_t-i}(t_{i+1} - t_i, \dots, t_{n_t} - t_i)$$

In the proof of Theorem 4, we conditioned the likelihood function by the last perfect repair time since censorship time (6). We can also condition by the first perfect repair time from 0. We can then prove the following recursive formula for the likelihood function.

Corollary 6

For a single observation of the failure process over $[0, t]$ (t may or may not be a failure time), the likelihood function verifies:

$$L_t(\theta) = \left[\sum_{j=1}^{n_t} p(1-p)^{j-1} \left[\prod_{i=1}^j \lambda(t_i) \right] e^{-\Lambda(t_j)} L_{t_j,t}(\theta) \right] + (1-p)^{n_t} \left[\prod_{i=1}^{n_t} \lambda(t_i) \right] e^{-\Lambda(t)}$$

Proof

$$L_t(\theta) = \left[\sum_{j=1}^{n_t} f_{\underline{T}_{n_t}, N_t=n_t | B_j=1, \{B_k=0\}_{0 < k < j}}(\underline{t}_{n_t}) P(B_j=1, \{B_k=0\}_{0 < k < j}) \right] + f_{\underline{T}_{n_t}, N_t=n_t | \{B_k=0\}_{0 < k \leq n_t}}(\underline{t}_{n_t}) P(\{B_k=0\}_{0 < k \leq n_t})$$

The end of the proof is quite similar to the proof of Theorem 4. $\square$

Langseth and Lindqvist [12] propose a likelihood function for a similar case, but they made a mistake. Before conditioning by $B_{N_t}^j$, they wrote the joint density of the failure process as a product of the conditional densities:

$$\begin{aligned} f_{\underline{T}_n}(\underline{t}_n) &= \prod_{i=1}^n f_{T_i | \underline{T}_{i-1} = \underline{t}_{i-1}}(t_i) \\ &= \prod_{i=1}^n \sum_{j=0}^{i-1} f_{T_i | \underline{T}_{i-1} = \underline{t}_{i-1}, B_{i-1}^j}(t_i) P(B_{i-1}^j | \underline{T}_{i-1} = \underline{t}_{i-1}) \end{aligned}$$

Furthermore, they considered B_j and T_i to be independent, i.e. $P(B_i^j | \underline{T}_i) = p^{\mathbb{1}_{\{j < i\}}}(1-p)^j$. However, B_j and T_i are independent only for $i \leq j$. In fact, if the last repair is perfect, for example, the next failure time will occur later than if the last repair is minimal.

4.3. Failure intensity

For a self-excited point process, the likelihood of a single observation of the failure process up to time $t \geq 0$ verifies (1). Failure intensity can be deduced from the likelihood function. For the BP model, failure intensity, like the likelihood function, is defined by a recursive formula. The following theorem provides the values of failure intensity except for failure times. However, failure intensity can easily be extended to $[0, \infty[$ because it is assumed to be left-continuous.

Theorem 5

$$\forall t \neq T_{N_t}, \quad \lambda_t = \frac{-d}{dt} \left[\ln \left(\sum_{j=0}^{N_t-} p^{\mathbb{1}_{\{j \neq 0\}}}(1-p)^{N_t-j} \left[\prod_{i=j+1}^{N_t-} \frac{\lambda(T_i - T_j)}{\lambda T_i} \right] e^{-\Lambda(t-T_j) - \Lambda T_j} \right) \right]$$

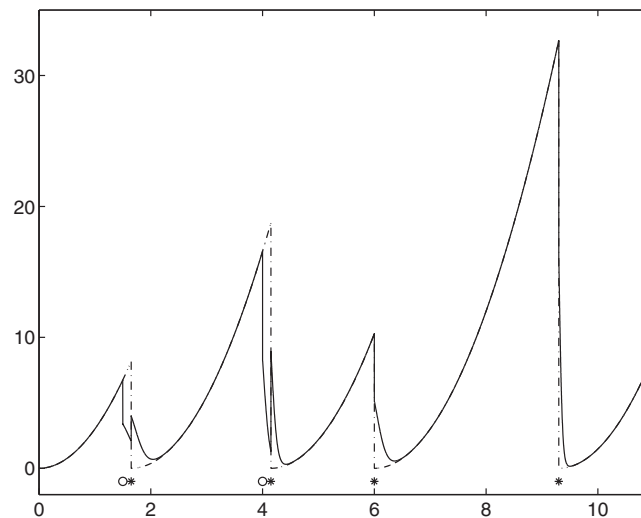


Figure 4. Unconditional and conditional failure intensities for $p = 0.5$.

Proof

The likelihood function with censorship at time T_j is equal to $[\prod_{n=1}^j \lambda_{T_n}] e^{-\Lambda_{T_j}}$. Then, using Theorem 4, the likelihood with censorship at time t is equal to:

$$\left[\prod_{n=1}^{N_t} \lambda_{T_n} \right] e^{-\Lambda_t} = \sum_{j=0}^{N_t-1} p^{\mathbb{1}_{\{j \neq 0\}}} (1-p)^{N_t-j} \left[\prod_{i=j+1}^{N_t} \lambda(T_i - T_j) \right] e^{-\Lambda(t-T_j)} \left[\prod_{n=1}^j \lambda_{T_n} \right] e^{-\Lambda_{T_j}}$$

Both sides of the equation were divided by $\prod_{n=1}^{N_t} \lambda_{T_n}$

$$e^{-\Lambda_t} = \sum_{j=0}^{N_t-1} p^{\mathbb{1}_{\{j \neq 0\}}} (1-p)^{N_t-j} \left[\prod_{i=j+1}^{N_t} \frac{\lambda(T_i - T_j)}{\lambda_{T_i}} \right] e^{-\Lambda(t-T_j) - \Lambda_{T_j}}$$

This equation can be differentiated for all $t > 0$, except for failure times. Finally, the theorem is proved by differentiating minus the logarithm of both sides of the equation. $\square$

Figure 4 shows the unconditional (solid line) and the conditional (dashed line) failure intensities for $p = 0.5$ for a simulated data set. The initial intensity is $\lambda(t) = 3t^2$. The stars and circles represent AGAN and ABAO repair times, respectively. After a repair, the failure intensity also seems to follow a bathtub curve. If the time elapsed between repairs is sufficiently long, failure intensity becomes equal to initial intensity at time $t - T_{N_t-}$. After a repair, the shape of failure intensity during what can be called ‘the burn in period’ depends on previous failure times. If the previous time between failures is long enough, the burn-in period seems to be less noticeable. These results are similar to those found for Theorem 2.

5. ML parameters’ estimators

This section proposes two efficient algorithms for likelihood computation. The EM method is derived from these algorithms. We also propose MLEs using a moment estimation of the scale parameter.

5.1. Direct likelihood maximization

For $k \geq 1$ and $j \in \{0, \dots, k-1\}$, let $P_k[j] = \prod_{i=j+1}^k \lambda(T_i - T_j)$. We notice that:

$$P_k[j] = \begin{cases} \lambda(T_k - T_j) P_{k-1}[j] & \text{for } 0 \leq j \leq k-2 \\ \lambda(T_k - T_{k-1}) & \text{for } j = k-1 \end{cases}$$

Then, using Theorem 4, the likelihood function can be computed with the following algorithm. This algorithm is called a forward algorithm since the likelihood function is recursively computed on time interval $[0, T_1]$, $\dots$, $[0, T_{N_t}]$ and finally $[0, t]$. Comments are written between $/**$ and $*/$.

Likelihood forward computation algorithm:

```

 $L_{T_0} = 1;$ 
for  $k$  from 1 to  $N_t$ 
   $P[k-1] = \lambda(T_k - T_{k-1}); /** = P_k[k-1]** /$ 
   $L_{T_k} = p^{\mathbb{1}_{\{k-1 \neq 0\}}} P[k-1] \exp(-\Lambda(T_k - T_{k-1})) L_{T_{k-1}};$ 
  for  $j$  from 0 to  $k-2$ 
     $P[j] = \lambda(T_k - T_j) P[j]; /** = P_k[j] \text{ with } j \leq k-2** /$ 
     $L_{T_k} = L_{T_k} + p^{\mathbb{1}_{\{j \neq 0\}}} (1-p)^{k-1-j} P[j] \exp(-\Lambda(T_k - T_j)) L_{T_j};$ 
  end for;
end for;
if  $T_{N_t} < t$ 
   $P[N_t] = 1;$ 
   $L_t = \sum_{j=0}^{N_t} p^{\mathbb{1}_{\{j \neq 0\}}} (1-p)^{N_t-j} P[j] \exp(-\Lambda(t - T_j)) L_{T_j};$ 
end if

```

The computing complexity of the likelihood computation is $O(1 + 2 + \dots + n) = O(n(n+1)/2) = O(n^2)$, where n is the number of failures observed. This algorithm can also be adapted in order to calculate the partial derivatives in θ of the likelihood function and its complexity remains the same. Then, the MLEs can be found numerically with any maximization method. In the examples below, the log-likelihood has been maximized with Matlab's `fmincon` function, again using the gradient of the likelihood.

A similar algorithm is developed from Corollary 6. This is a backward computation algorithm since likelihood is recursively computed on time interval $[t - T_{N_t}, t], \dots, [T_0, t]$.

Likelihood backward computation algorithm:

```

 $L_{T_{N_t}, t} = \exp(-\Lambda(t - T_{N_t}));$ 
for  $k$  from  $N_t$  to 1 by  $-1$ 
   $P = 1; L_{T_{k-1}, t} = 0;$ 
  for  $j$  from  $k$  to  $N_t$ 
     $P = \lambda(T_j - T_{k-1}) P;$ 
     $L_{T_{k-1}, t} = L_{T_{k-1}, t} + p(1-p)^{j-k} P \exp(-\Lambda(T_j - T_{k-1})) L_{T_j, t};$ 
  end for;
   $L_{T_{k-1}, t} = L_{T_{k-1}, t} + (1-p)^{N_t-k+1} P \exp(-\Lambda(t - T_{k-1}));$ 
end for;
 $L_t = L_{T_0, t}$ 

```

5.2. EM algorithm

When a model is naturally defined with hidden variables, the likelihood function is often intractable. In this case, the EM algorithm seeks to find the MLE by iteratively applying the following two steps:

- Expectation (E) step: Calculate the expected value of the complete log-likelihood function (associated with the observed (N) and hidden (B) variables), with respect to the conditional distribution of B given N under the current estimate of the parameters θ_k :

$$Q(\theta|\theta_k) = E^{\theta_k}[\log(L_t^c(\theta)|N)]$$

- Maximization (M) step: Find the parameters that maximize this quantity:

$$\theta_{k+1} = \arg \max_{\theta} Q(\theta|\theta_k)$$

Lim [9] applied the EM algorithm to search for BP model estimators (with a Weibull-distributed first time to failure) when repair effects $\{B_i\}_{i \geq 1}$ are unknown. In this section, we describe how the EM algorithm can be derived from the results already developed in this paper.

The complete likelihood associated with the observation of failure times and repair effects is given by [3]:

$$L_t^c(\theta) = \left[\prod_{i=1}^{n_t} p^{B_i} (1-p)^{1-B_i} \lambda_{t_i}(N, B) \exp\left(-\int_{t_{i-1}}^{t_i} \lambda_s(N, B) ds\right) \right] \exp\left(-\int_{t_{n_t}}^t \lambda_s(N, B) ds\right)$$

And the conditional failure intensity verifies:

$$\forall s \in]t_{i-1}, t_i], \quad \lambda_s(N, B) = \prod_{j=0}^{i-1} [\lambda(s - t_{i-1-j})]^{\mathbb{1}_{\{B_{i-1}^j\}}}$$

Let us denote $\pi_{i,j}^{\theta_k} = E^{\theta_k}[\mathbb{1}_{\{B_i^j\}} | N_t = n_t, \underline{T}_{n_t} = \underline{t}_{n_t}]$. It can be observed that $\pi_{i,0}^{\theta_k} = E^{\theta_k}[B_i | N_t = n_t, \underline{T}_{n_t} = \underline{t}_{n_t}]$.

Then, the expected value of the complete log-likelihood verifies

$$Q(\theta | \theta_k) = Q_1(p | \theta_k) + Q_2(\lambda | \theta_k)$$

with

$$Q_1(p | \theta_k) = \log(p) \left(\sum_{i=1}^{n_t} \pi_{i,0}^{\theta_k} \right) + \log(1-p) \left[n_t - \left(\sum_{i=1}^{n_t} \pi_{i,0}^{\theta_k} \right) \right]$$

$$Q_2(\lambda | \theta_k) = \left(\sum_{i=1}^{n_t} \sum_{j=0}^{i-1} \pi_{i-1,j}^{\theta_k} [\log(\lambda(t_i - t_{i-1-j})) - (\Lambda(t_i - t_{i-1-j}) - \Lambda(t_{i-1} - t_{i-1-j}))] \right) \\ - \left(\sum_{j=0}^{n_t} \pi_{n_t,j}^{\theta_k} (\Lambda(t - t_{n_t-j}) - \Lambda(t_{n_t} - t_{n_t-j})) \right)$$

5.2.1. *E step.* Thus, the E step results in calculating $\pi_{i,j}^{\theta_k}$ for $0 \leq i \leq n_t$ and $0 \leq j \leq i$.

First, for $1 \leq i \leq n_t$

$$\pi_{i,0}^{\theta} = P^{\theta}(B_i = 1 | N_t = n_t, \underline{T}_{N_t} = \underline{t}_{N_t}) \\ = P^{\theta}(B_i = 1) \frac{f_{\underline{T}_{n_t}, N_t = n_t | B_i = 1}^{\theta}(\underline{t}_{n_t})}{f_{\underline{T}_{n_t}, N_t = n_t}^{\theta}(\underline{t}_{n_t})}$$

Given $B_i = 1$, the failure process was renewed in t_i . Then it can be divided into two independent identical BP failure processes. Hence,

$$\forall i \in \{1, \dots, n_t\}, \quad \pi_{i,0}^{\theta} = p \frac{L_{t_i}(\theta) L_{t_i, t}(\theta)}{L_t(\theta)} \quad \text{and} \quad \pi_{0,0}^{\theta} = 1 \quad (7)$$

Second, for $0 \leq i < n_t$ and $0 \leq j \leq i$,

$$\pi_{i,j}^{\theta} = P^{\theta}(B_i^j | N_t = n_t, \underline{T}_{N_t} = \underline{t}_{N_t}) \\ = P^{\theta}(B_i^j, B_{i+1} = 0 | N_t = n_t, \underline{T}_{N_t} = \underline{t}_{N_t}) + P^{\theta}(B_i^j, B_{i+1} = 1 | N_t = n_t, \underline{T}_{N_t} = \underline{t}_{N_t}) \\ = \pi_{i+1,j+1}^{\theta} + P^{\theta}(B_i^j, B_{i+1} = 1 | \underline{T}_{i-j} = \underline{t}_{i-j}) \frac{f_{T_{i-j+1}, \dots, T_{n_t}, N_t = n_t | \underline{T}_{i-j} = \underline{t}_{i-j}, B_i^j, B_{i+1} = 1}^{\theta}(t_{i-j+1}, \dots, t_{n_t})}{f_{T_{i-j+1}, \dots, T_{n_t}, N_t = n_t | \underline{T}_{i-j} = \underline{t}_{i-j}}^{\theta}(t_{i-j+1}, \dots, t_{n_t})}$$

First, the repair effect B_i is independent of previously observed repair times, T_k for $k \leq i$. Then $P^{\theta}(B_i^j, B_{i+1} = 1 | \underline{T}_{i-j} = \underline{t}_{i-j}) = P^{\theta}(B_i^j, B_{i+1} = 1) = p^{1 + \mathbb{1}_{\{i > j\}}}(1-p)^j$. Second,

$$f_{T_{i-j+1}, \dots, T_{n_t}, N_t = n_t | \underline{T}_{i-j} = \underline{t}_{i-j}}^{\theta}(t_{i-j+1}, \dots, t_{n_t}) = \frac{f_{\underline{T}_{n_t}, N_t = n_t}^{\theta}(\underline{t}_{n_t})}{f_{\underline{T}_{i-j}}^{\theta}(\underline{t}_{i-j})} = \frac{L_t(\theta)}{L_{t_{i-j}}(\theta)}$$

Finally, given B_i^j and $B_{i+1} = 1$, the failure process was renewed in t_{i-j} and t_{i+1} . Then, it can be divided into three independent processes. Between t_{i-j} and t_{i+1} , all the repairs are assumed to be ABAO; therefore, failure times are those of an NHPP

initialized at t_{i-j} and with a failure intensity $\lambda(t)$. After t_{i+1} , the failure times are those of a BP model initialized at t_{i+1} . Then

$$f_{T_{i-j+1}, \dots, T_{n_t}, N_t=n_t | T_{i-j}=t_{i-j}, B_i^j, B_{i+1}=1}^\theta(t_{i-j+1}, \dots, t_{n_t}) = \left[\prod_{k=i-j+1}^{i+1} \lambda(t_k - t_{i-j}) \right] e^{-\Lambda(t_{i+1}-t_{i-j})} L_{t_{i+1}, t}(\theta)$$

In conclusion, the E step results in applying the likelihood forward and backward computation algorithms of Section 5.1 and then computing $\pi_{i,j}^\theta$ with (7) and recursively for all $0 \leq i < n_t$ and $0 \leq j \leq i$ with

$$\pi_{i+1,j+1}^\theta = \pi_{i,j}^\theta - p^{1+\mathbb{I}_{\{i>j\}}}(1-p)^j \left[\prod_{k=i-j+1}^{i+1} \lambda(t_k - t_{i-j}) \right] e^{-\Lambda(t_{i+1}-t_{i-j})} \frac{L_{t_{i+1}, t}(\theta) L_{t_{i-j}}(\theta)}{L_t(\theta)}$$

The computing complexity of the E step computation is equivalent to the complexity of the computation of the likelihood and its derivatives. The memory complexity of the E step computation is squared in comparison to the likelihood computation.

5.2.2. M step. The parameter p_{k+1} , which maximizes the expectation of the log-likelihood, maximizes $Q_1(p|\theta_k)$ and then causes its derivative to vanish. Hence, p_{k+1} verifies:

$$p_{k+1} = \frac{\sum_{i=1}^{n_t} \pi_{i,0}^{\theta_k}}{n_t} \quad (8)$$

This is quite a natural estimator of repair efficiency since it also verifies:

$$p_{k+1} = E^{\theta_k} \left[\frac{\sum_{i=1}^{n_t} B_i}{n_t} \middle| N_t = n_t, T_{n_t} = t_{n_t} \right]$$

The initial intensity parameters, which maximize the expectation of the log-likelihood, maximize $Q_2(\lambda|\theta_k)$ and cause its derivatives to vanish. Q_2 has the same form as a weighted likelihood for an NHPP with a failure intensity equal to initial intensity. Then, if the first time to failure is Weibull-distributed (2), η_{k+1} verifies:

$$\eta_{k+1} = \left(\frac{S_k(\beta_{k+1})}{\tilde{S}_k} \right)^{1/\beta_{k+1}} = \left(\frac{S_k(\beta_{k+1})}{n_t} \right)^{1/\beta_{k+1}} \quad (9)$$

with $\tilde{S}_k = \sum_{i=1}^{n_t} \sum_{j=0}^{i-1} \pi_{i-1,j}^{\theta_k} = \sum_{i=1}^{n_t} 1 = n_t$ and

$$S_k(\beta) = \left[\sum_{i=1}^{n_t} \sum_{j=0}^{i-1} \pi_{i-1,j}^{\theta_k} ((t_i - t_{i-1-j})^\beta - (t_{i-1} - t_{i-1-j})^\beta) \right] + \left[\sum_{j=0}^{n_t} \pi_{n_t,j}^{\theta_k} ((t - t_{n_t-j})^\beta - (t_{n_t} - t_{n_t-j})^\beta) \right]$$

In conclusion, the E step is quite simple because it results in maximizing a one-dimensional equation

$$\beta_{k+1} = \arg \max_{\beta} \left[n_t \left(\log \left(\frac{n_t \beta}{S_k(\beta)} \right) \right) + (\beta - 1) \left[\sum_{i=1}^{n_t} \sum_{j=0}^{i-1} \pi_{i-1,j}^{\theta_k} \log(t_i - t_{i-1-j}) \right] \right] \quad (10)$$

and computing two parameters (8) and (9).

5.3. The scale parameter

In the following, the first time to failure is assumed to be Weibull-distributed (2).

5.3.1. Moment estimation of the scale parameter. Traditionally, for power law processes (PLPs) or Weibull RP, the scale parameter MLE $\hat{\eta}$ has a closed-form solution function of the shape parameter MLE. This is not the case for the BP model. Hence, the initialization value of the scale parameter has to be chosen carefully in order to allow numeric computation of the likelihood function. A good initialization of the scale parameter can be obtained using Corollary 5. In fact it implies:

$$E \left[\frac{1}{n} \sum_{i=1}^n X_i \right] = \frac{\eta}{\beta} \left[\sum_{i=0}^{n-1} \frac{1 + p(n-i-1)}{n} (1-p)^i \frac{\Gamma(i+1/\beta)}{\Gamma(i+1)} \right] \xrightarrow[p \rightarrow \infty]{p > 0} \frac{p^{(\beta-1)/\beta} \eta}{\beta} \Gamma \left(\frac{1}{\beta} \right)$$

Then, the moment estimator of the scale parameter verifies:

$$\tilde{\eta}_{\beta,p}(n) = \frac{\beta T_n}{\sum_{i=0}^{n-1} (1+p(n-i-1))(1-p)^i \frac{\Gamma(i+1/\beta)}{\Gamma(i+1)}} \stackrel{p>0}{\sim} \frac{\beta T_n}{n p^{(\beta-1)/\beta} \Gamma(1/\beta)}$$

If the true values of β and p are known, the estimator of the scale parameter is unbiased, whatever be the sample size. But more generally, for any value of β and p , this estimator is a good approximation of MLE.

$$\tilde{\eta}_{\beta,p}(n_t) \approx \arg \max_{\eta} \log(L_t(\eta, \beta, p))$$

Then, using this moment estimator we can reduce the dimension space of the likelihood maximization:

$$(\tilde{\beta}, \tilde{p}) = \arg \max_{(\beta,p)} \log(L_t(\tilde{\eta}_{\beta,p}(n_T), \beta, p)) \quad \text{and} \quad \tilde{\eta} = \tilde{\eta}_{\tilde{\beta},\tilde{p}}(n_t)$$

These estimators are close to the MLEs $\tilde{\theta} \approx \hat{\theta}$. They can be used to estimate the parameters directly or to initialize the ML method.

5.3.2. A scale parameter for initial intensity and for the BP model. The following properties can easily be proved from previous results. They imply that the scale parameter of the first time to failure distribution η is also a scale parameter for the BP model with unknown repair effects.

For any constant $c > 0$, let us define the BP likelihood function associated with $c t_1, \dots, c t_{n_t}, n_t$ (the homothetic transformation of the observation by c):

$$L_{ct}(\theta) = f_{\underline{T}_n, N_{ct}=n_t}(c \underline{T}_n)$$

The following proposition is a direct outcome of Theorem 4.

Proposition 1

$\forall c > 0, \forall \theta = (\eta, \beta, p) \in]0, \infty[\times [1, \infty[\times [0, 1], \forall t > 0$

$$L_t(\theta) = \frac{1}{c^{n_t}} L_{ct}(\theta') \quad \text{with } \theta' = (c\eta, \beta, p)$$

Following corollary can be deduced directly from this proposition.

Corollary 7

$$F_{X_n|\underline{T}_{n-1}=\underline{t}_{n-1}}^{\theta}(x) = F_{X_n|\underline{T}_{n-1}=c\underline{t}_{n-1}}^{\theta'}(cx)$$

And if $F_{X_n|\underline{T}_{n-1}=\underline{t}_{n-1}}^{\theta}$ is invertible then $F_{X_n|\underline{T}_{n-1}=c\underline{t}_{n-1}}^{\theta'}$ is also invertible and

$$(F_{X_n|\underline{T}_{n-1}=\underline{t}_{n-1}}^{\theta})^{-1}(x) = \frac{1}{c} (F_{X_n|\underline{T}_{n-1}=c\underline{t}_{n-1}}^{\theta'})^{-1}(x)$$

Then, η is a scale parameter for the BP model.

Corollary 8

For all $n \geq 1$, if $\underline{T}_n$ follows a BP model with parameter $\theta = (\eta, \beta, p)$, then $c \underline{T}_n$ follows a BP model with parameter $\theta' = (c\eta, \beta, p)$.

The following property is also a direct outcome of Proposition 1.

Corollary 9

If a unique $\hat{\theta}$ exists such that $\hat{\theta} = \arg \max_{\theta} L_t(\theta)$, then

$$\arg \max_{\theta} L_{ct}(\theta) = \hat{\theta}' = (c\hat{\eta}, \hat{\beta}, \hat{p})$$

Then, the law of the MLEs $\hat{\beta}$ and $\hat{p}$ of the shape and repair efficiency parameters do not depend on the real value of the scale parameter η . Similarly, the same property holds for estimators $\tilde{\beta}$ and $\tilde{p}$.

6. Application

In the following, the first time to failure is assumed to be Weibull-distributed (2).

6.1. Numerical application

Let us consider the data set for the hydraulic systems of LHD machines given by Kumar and Klefsjö [13]. It consists of times between failures collected over 2 years, from three different machines groups representing old (LHD 1 and LHD 3), medium old (LHD 9 and LHD 11) and new machines (LHD 17 and LHD 20). Using classical statistical methods, Kumar and Klefsjö highlighted that the times between failures are not iid but could be described with a PLP with an increasing shape parameter. Contrary to expectations, no significant growth in reliability was noticed between new and older machines. This could be due to repair effects that compensate for wear-out. We propose to use the BP model with this data set.

We focused on LHD 11 since it is the machine with the most failure times: 28. However, similar results can be obtained for other machines.

The MLE for the traditional PLP model used by Kumar and Klefsjö (equivalent to the BP model with $p=0$, i.e. ABAO repairs) verifies:

$$\hat{\eta}=232, \quad \hat{\beta}=1.32, \quad \log(L_t(\hat{\eta}, \hat{\beta}, 0))=-157.09$$

Direct likelihood maximization and EM algorithm for the BP model provide the same estimates:

$$\hat{\eta}=299, \quad \hat{\beta}=3.56, \quad \hat{p}=0.226, \quad \log(L_t(\hat{\eta}, \hat{\beta}, \hat{p}))=-152.86$$

MLE with moment estimation of the scale parameter is also very similar:

$$\tilde{\eta}=299, \quad \tilde{\beta}=3.57, \quad \tilde{p}=0.239, \quad \log(L_t(\tilde{\eta}, \tilde{\beta}, \tilde{p}))=-152.87$$

Bayesian Information Criterion (BIC), Akaike's Information Criterion (AIC), Akaike's Information Criterion with a second order correction for small sample sizes (AICc) are criteria commonly used for model selection.

- For the PLP model, BIC=320.84, AIC=318.18, AICc=318.34.
- For the BP model, BIC=315.71, AIC=311.72, AICc=312.22.

The BP model should therefore be recommended for this data set, since it has the lowest criteria values.

The estimated values of β greater than 3 indicate a strong intrinsic wear-out. But the system is renewed by 23% of repairs. Then, the overall repair actions compensate for wear-out.

Figure 5 represents the failure intensity of the BP model for the estimated parameter value. Stars represent failure times. Between failures, failure intensity follows a bathtub curve. When time between failures becomes sufficiently long, failure intensity is almost renewed. Conversely, in the case of failure surge, failure intensity increases significantly. Then, the system seems to wear-out, but is regularly renewed by repair actions, so that wear-out is compensated by repair actions.

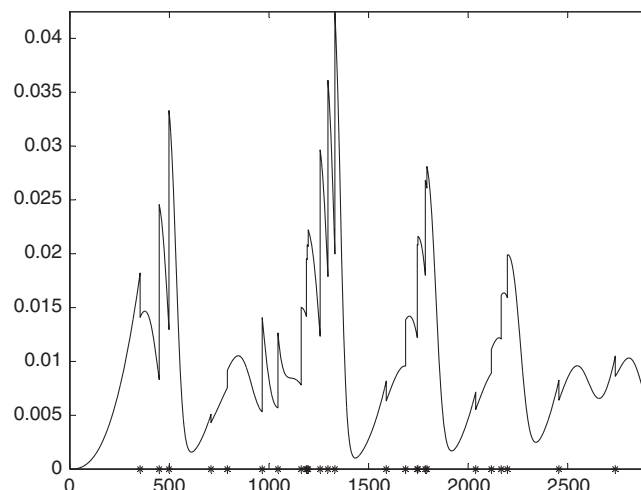


Figure 5. Failure intensity of LHD 11.

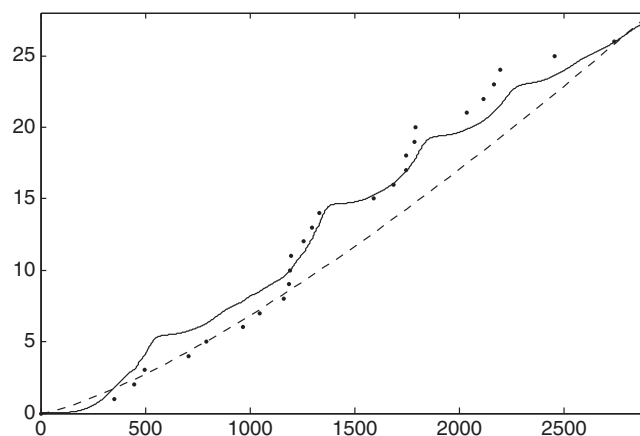


Figure 6. Cumulative failure intensity of LHD 11.

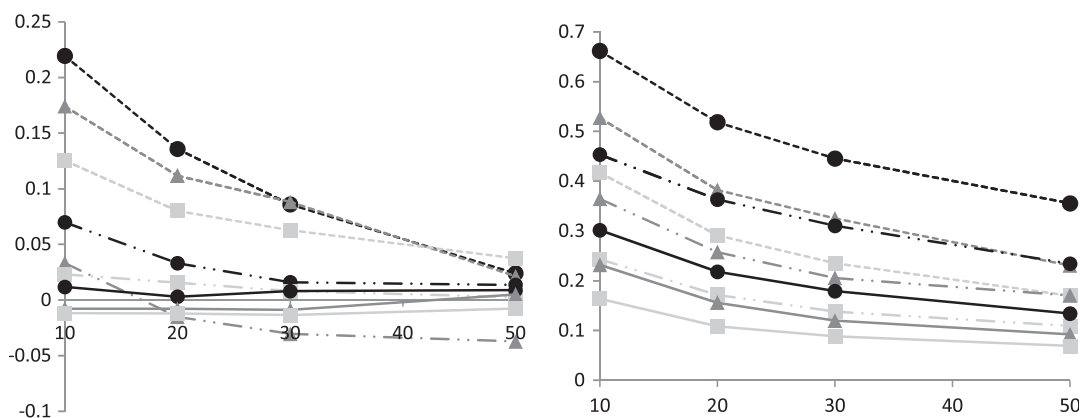


Figure 7. Normalized empirical bias (left) and standard deviation (right) of $\hat{\eta}$.

Figure 6 represents the cumulative failure intensities of the BP model (solid line) and PLP model (dashed line) for estimated parameters values. Points represent the cumulative number of failures. The BP model seems to fit the data overall.

6.2. Empirical study of estimators' quality

In order to study the quality of the estimators, we computed (Figures 7–9) the Normalized Empirical Bias, NEB, (that is to say the empirical bias divided by the real parameter value) and Standard Deviation, NESD, (that is to say the empirical standard deviation divided by the real parameter value) function of sample size ($n \in \{10, 20, 30, 50\}$ failure times) over 1000 simulations for:

- $\eta=1, \beta=1.5, p=0.2$ --▲-- $\eta=1, \beta=1.5, p=0.5$ --■-- $\eta=1, \beta=1.5, p=0.8$
- $\eta=1, \beta=2, p=0.2$ —▲— $\eta=1, \beta=2, p=0.5$ —■— $\eta=1, \beta=2, p=0.8$
- $\eta=1, \beta=3, p=0.2$ —▲— $\eta=1, \beta=3, p=0.5$ —■— $\eta=1, \beta=3, p=0.8$

First we noticed that NEB and NESD of MLE computed with the direct likelihood maximization or EM algorithm, and of MLE with moment estimation of the scale parameter ($\tilde{\theta}$), are the same with respect to empirical bias and standard deviation precision. Therefore, the $\tilde{\theta}$ estimator appears preferable since it is easier to compute.

Second, we noticed that the bias and standard deviation of every estimator decrease naturally as the sample size increases. NEB and NESD of all estimators globally decrease as the value of the shape or repair efficiency parameter increases: the quality of the estimation is generally better for wear-out systems with efficient repair. The worst estimated parameter is the repair efficiency parameter in the case of poor repair efficiency (low p) associated with weak wear-out (β close to 1). In this case, the quality of the repair efficiency parameter remains low even if the sample increases. The shape parameter is overestimated on average in all cases.

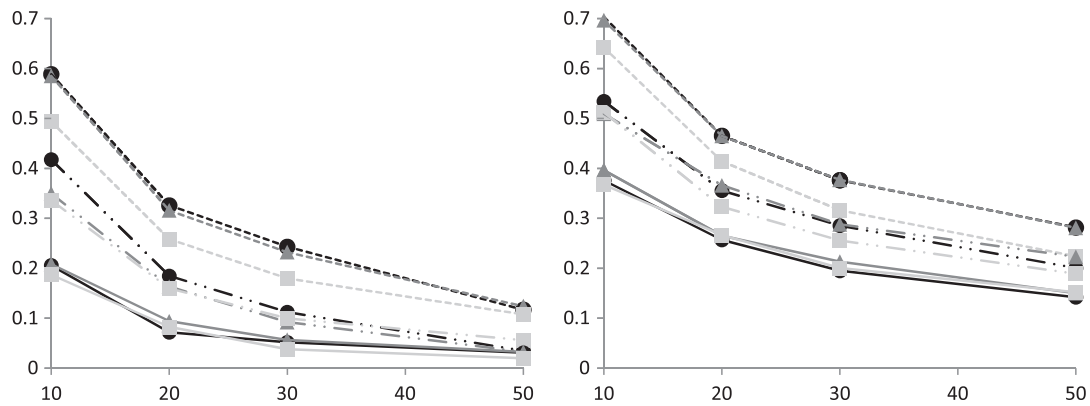


Figure 8. Normalized empirical bias (left) and standard deviation (right) of $\hat{\beta}$.

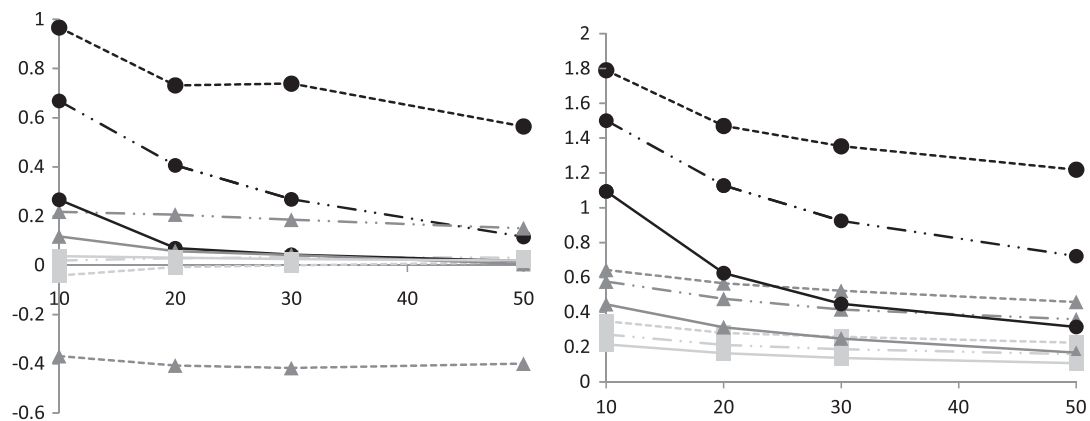


Figure 9. Normalized empirical bias (left) and standard deviation (right) of $\hat{p}$.

7. Conclusion

This paper provides a framework for assessing repair efficiency and intrinsic wear-out. However, critical systems are generally preventively maintained. The main results developed in this paper could certainly be generalized to the case of systems subject to CM and PM.

Here, we have highlighted the advantage of considering a model with unknown random effects, but in the BP model, the random effect is simply assumed to be Bernoulli-distributed. A natural extension of the model would be to consider that B_i has a more general continuous distribution over $]-\infty, 1]$. Then it will be possible to consider not only AGAN ($B_i = 1$) and ABAO ($B_i = 0$) repairs, but any efficient imperfect repair ($0 < B_i < 1$) and even damaging repair ($B_i < 0$). This is equivalent to consider a Kijima's type II model with unknown random repair effects.

Most of the papers studying statistical properties of the BP model propose non-parametric estimators of the distribution of time to first failure when repair effects are known [3–5, 8]. Then, a prospect could be to develop semi-parametric methods in the case of unknown repair effects.

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